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Inventor(s)	Snyder; Todd C.

Dental bur

Abstract

A dental bur for removing precise portions of a tooth includes a shank and a head. The shank includes first and second end, an attachment section located adjacent the first end, and a guide section located between the attachment section and the second end. The attachment section is configured to be inserted into and rotationally driven by a high-speed rotary device. The guide section includes at least one portion having a first width measured in a transverse direction of the shank and a second portion having a second width measured in the transverse direction. The first width being different than the second width. The head is arranged on the second end of the shank and includes a tip located opposite the shank. The tip has a rounded portion configured to remove portions of the tooth, and the head is entirely coated in an abrasive diamond material.

Inventors:	Snyder; Todd C. (Laguna Niguel, CA)
Applicant:	Snyder; Todd C. (Laguna Niguel, CA)
Family ID:	84799505
Assignee:	Snyder; Todd C. (Laguna Niguel, CA)
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Primary Examiner: Rosen; Eric J

Assistant Examiner: Mai; Hao D

Attorney, Agent or Firm: Barnes & Thornburg LLP

Background/Summary

BACKGROUND

(1) The present disclosure relates to dental tools. More particularly, the present disclosure relates to dental burs for high-speed dental drills.

SUMMARY

(2) According to the present disclosure, a dental bur for removing precise portions of a tooth includes a shank and a head. The shank has a first terminal end, a second terminal end located opposite the first terminal end in a longitudinal direction of the shank, an attachment section located adjacent the first terminal end, and a guide section located between the attachment section and the second terminal end, the attachment section being configured to inserted into and rotationally driven by a high-speed rotary device, the guide section including at least one portion having a first width measured in a transverse direction of the shank and a second portion having a second width measured in the transverse direction, and the first width being different than the second width. The head is arranged on the second terminal end of the shank and including a tip located opposite the shank, the tip having at least one rounded portion configured to remove portions of the tooth, and the head being entirely coated in an abrasive diamond material.

(3) In at least some embodiments, the guide section includes an inner end located adjacent the attachment section and an outer end located adjacent the head, the at least one portion having the first width is located at the inner end of the guide section, the second portion having the second width is located at the outer end of the guide section, and the first width is greater than the second width.

(4) In at least some embodiments, the guide section has a constant first taper from the at least one portion having the first width to the second portion having the second width so as to form a frustum-shape.

(5) In at least some embodiments, the at least one rounded portion of the tip has a semi-spherical shape.

(6) In at least some embodiments, the head includes an inner terminal end located adjacent the second terminal end of the shank and a main body portion located between the inner terminal end and the tip, and the inner terminal end has a third width that is equal to the second width of the second portion of the guide section of the shank.

(7) In at least some embodiments, an inner end of the tip located adjacent the main body portion of the head has a fourth width that is less than the third width.

(8) In at least some embodiments, the main body portion of the head has a constant second taper from the inner terminal end having the third width to the inner end of the tip having the fourth width so as to form a frustum-shape, and a taper angle of the second taper is equal to a taper angle of the first taper of the guide section of the shank.

(9) In at least some embodiments, the guide section of the shank and the main body portion of the head are generally cylindrical such that the first, second, third, and fourth widths are diameters of the generally cylindrical shank and head.

(10) In at least some embodiments, the head includes an inner terminal end located adjacent the second terminal end of the shank, a main body portion located between the inner terminal end and the tip, and an inner end of the tip located adjacent the main body portion of the head, wherein the at least one rounded portion of the tip is an annular rounded ridge that extends around the tip at the inner end of the tip, and wherein a distal end of the tip located opposite the inner end of the tip is an annular flat surface connected to the annular rounded ridge.

(11) In at least some embodiments, the guide section includes an inner end located adjacent the attachment section, an outer end located adjacent the head, and an intermediate section located between the inner end and the outer end, the at least one portion having the first width is located at the inner end of the guide section, the second portion having the second width is located at the outer end of the guide section, a third portion of the guide section is located between the at least

one portion and the second portion within the intermediate section and has a third width measured in the transverse direction, and the first width and the second width are greater than the third width such that the guide section has an hourglass shape.

(12) In at least some embodiments, the guide section of the shank is generally cylindrical such that the first, second, and third widths are diameters of the generally cylindrical shank.

(13) According to another aspect of the present disclosure, a high-speed dental drill includes a drill head, a bur mount, a motor, and a dental bur. The bur mount is arranged within the drill head and configured to rotate at high speeds, and the motor is for rotating the bur mount. The dental bur is arranged within the bur mount and includes a shank and a head.

(14) The shank has a first terminal end, a second terminal end located opposite the first terminal end in a longitudinal direction of the shank, an attachment section located adjacent the first terminal end, and a guide section located between the attachment section and the second terminal end, the attachment section being configured to inserted into and rotationally driven by a high-speed rotary device, the guide section including at least one portion having a first width measured in a transverse direction of the shank and a second portion having a second width measured in the transverse direction, and the first width being different than the second width.

(15) The head is arranged on the second terminal end of the shank and including a tip located opposite the shank, the tip having at least one rounded portion configured to remove portions of the tooth, and the head being entirely coated in an abrasive diamond material.

(16) In at least some embodiments, the guide section includes an inner end located adjacent the attachment section and an outer end located adjacent the head, the at least one portion having the first width is located at the inner end of the guide section, the second portion having the second width is located at the outer end of the guide section, and the first width is greater than the second width.

(17) In at least some embodiments, the guide section has a constant first taper from the at least one portion having the first width to the second portion having the second width so as to form a frustum-shape.

(18) In at least some embodiments, the at least one rounded portion of the tip has a semi-spherical shape.

(19) In at least some embodiments, the head includes an inner terminal end located adjacent the second terminal end of the shank and a main body portion located between the inner terminal end and the tip, and the inner terminal end has a third width that is equal to the second width of the second portion of the guide section of the shank.

(20) In at least some embodiments, the main body portion of the head has a constant second taper from the inner terminal end having the third width to the inner end of the tip having the fourth width so as to form a frustum-shape, and a taper angle of the second taper is equal to a taper angle of the first taper of the guide section of the shank.

(21) In at least some embodiments, the head includes an inner terminal end located adjacent the second terminal end of the shank, a main body portion located between the inner terminal end and the tip, and an inner end of the tip located adjacent the main body portion of the head, wherein the at least one rounded portion of the tip is an annular rounded ridge that extends around the tip at the inner end of the tip, and wherein a distal end of the tip located opposite the inner end of the tip is an annular flat surface connected to the annular rounded ridge

(22) In at least some embodiments, the guide section includes an inner end located adjacent the attachment section, an outer end located adjacent the head, and an intermediate section located between the inner end and the outer end, the at least one portion having the first width is located at the inner end of the guide section, the second portion having the second width is located at the outer end of the guide section, a third portion of the guide section is located between the at least one portion and the second portion within the intermediate section and has a third width measured in the transverse direction, and the first width and the second width are greater than the third width

such that the guide section has an hourglass shape.

(23) Additional features of the present disclosure will become apparent to those skilled in the art upon consideration of illustrative embodiments exemplifying the best mode of carrying out the disclosure as presently perceived.

Description

BRIEF DESCRIPTIONS OF THE DRAWINGS

(1) The detailed description particularly refers to the accompanying figures in which:

(2) FIG. 1 is a front elevation view of a dental bur according to the present disclosure, showing that the dental bur includes a shank and a head having an abrasive diamond material applied thereto, the head having a semi-spherical shape, and the shank including a guide section having a constant taper and not having an abrasive material applied thereto so as to allow for precision application of the head to a tooth for minimal and precise tooth material removal without the guide section removing tooth material when coming into contact with adjacent teeth;

(3) FIG. 2 is a perspective view of the dental bur of FIG. 1, showing the dental bur attached to a dental drill including a head having a bur mount, and showing the dental bur inserted into a lower area of a space between two adjacent teeth in order to precisely remove tooth material;

(4) FIG. 3 is a front elevation view of the head of the dental bur of FIG. 1, showing the tip having a semi-spherical shape, and showing that the head includes an inner terminal end located adjacent a second terminal end of the shank and a main body portion located between the inner terminal end and the tip, and suggesting that the inner terminal end has a width that is equal to the width of the second terminal end of the shank;

(5) FIG. 4 is a side view of the dental bur as arranged within the dental tool of FIGS. 1 and 2, showing the dental bur attached to the dental drill;

(6) FIG. 5 is a perspective view of the dental bur as arranged within the dental tool of FIGS. 1 and 2, showing an additional view of the dental bur inserted into a lower area of a space between two adjacent teeth in order to precisely remove tooth material;

(7) FIG. 6 is a perspective view similar to FIG. 5 of the dental bur as arranged within the dental tool of FIGS. 1 and 2, showing an additional view of the dental bur inserted into a lower area of a space between two adjacent teeth in order to precisely remove tooth material;

(8) FIG. 7 is a side view of the dental bur as arranged within the dental tool of FIGS. 1 and 2, showing an additional view of the dental bur inserted from the side of the teeth into a lower area of a space between two adjacent teeth in order to precisely remove tooth material;

(9) FIG. 8 is a front elevation view of the dental bur of FIG. 1, showing the dental bur inserted into a lower area of a space between two adjacent teeth in order to precisely remove tooth material;

(10) FIG. 9 is a front elevation view of a dental bur according to another embodiment of the present disclosure, showing that the dental bur includes a shank and a head having an abrasive diamond material applied thereto, the head having a chamfer shape, and the shank including a guide section having a constant taper and not having an abrasive material applied thereto so as to allow for precision application of the head to a tooth for minimal and precise tooth material removal without the guide section removing tooth material when coming into contact with adjacent teeth;

(11) FIG. 10 is a front elevation view of the head of the dental bur of FIG. 9, showing the tip having a chamfer shape, and showing that the head includes an inner terminal end located adjacent a second terminal end of the shank and a main body portion located between the inner terminal end and the tip, and suggesting that the inner terminal end has a width that is equal to the width of the second terminal end of the shank;

(12) FIG. 11 is a front elevation view of a dental bur according to another embodiment of the present disclosure, showing that the dental bur includes a shank and a head having an abrasive

diamond material applied thereto, the head having a chamfer shape, and the shank including a guide section having an hourglass shape and not having an abrasive material applied thereto so as to allow for precision application of the head to a tooth for minimal and precise tooth material removal without the guide section removing tooth material when coming into contact with adjacent teeth; and

(13) FIG. 12 is a front elevation view of a dental bur according to another embodiment of the present disclosure, showing that the dental bur includes a shank and a head having an abrasive diamond material applied thereto, the head having a chamfer shape, and the shank including a guide section having a straight, cylindrical shape and not having an abrasive material applied thereto so as to allow for precision application of the head to a tooth for minimal and precise tooth material removal without the guide section removing tooth material when coming into contact with adjacent teeth.

DETAILED DESCRIPTION

(14) A first embodiment of a dental bur **10** in accordance with the present disclosure is shown in FIGS. 1-8. In one example, the dental bur **10** includes a shank **12** having an attachment section **14** and a guide section **20**, and a head **30** arranged on a terminal end **18** of the shank **12**. The head **30** of the dental bur **10** includes a tip **32** having at least one rounded portion configured to precisely remove portions of a tooth during preparation of a tooth or multiple teeth for application of a crown, veneers, or similar procedure without damaging adjacent teeth or restoration. In the illustrative embodiment, the head **30** is entirely coated in an abrasive diamond material **35** which is capable of removing portions of an outer surface of the tooth when the bur **10** is rotated at high-speeds when arranged within a high-speed drill **80**. The shank **12**, however, is not coated in any type of abrasive material. This allows for greater precision in the tooth tissue removal due to the shank **12** not being able to remove any tooth tissue that it may accidentally come into contact with when a user is focusing on particular areas of a tooth with the head **30** of the bur **10**. A second embodiment of a dental bur **110** according to the present disclosure is shown in FIGS. 9 and 10. Another embodiment of a dental bur **210** according to the present disclosure is shown in FIG. 11. Another embodiment of a dental bur **310** according to the present disclosure is shown in FIG. 12. (15) In the illustrative embodiment, the dental bur **10** of the present disclosure includes the shank **12** and the head **30**, as shown in FIGS. 1-8 and in greater detail in FIGS. 1, 3 and 8. The shank **12** of the dental bur **10** may be formed from metallic materials and is configured to be inserted into and arranged within a hand-held dental drill **80** such that a user may perform dental procedures on a patient's teeth using the dental bur **10**, in particular the diamond-coated head **30** of the dental bur **10**. In the illustrative embodiment, the shank **12** is shaped as a cylinder such that the dental bur **10** can be rotationally driven by the dental drill **80** at very high speeds without any lateral forces acting on the drill **80** and bur **10**.

(16) The shank **12** includes a first terminal end **17** and a second terminal end **18** located opposite the first terminal end **17** in a longitudinal direction of the shank **12**, as shown in FIG. 1. The shank **12** includes the attachment section **14** located adjacent the first terminal end **17** and a guide section **20** located between the attachment section **14** and the second terminal end **18** as can be seen in FIG. 1. The attachment section **14** of the shank **12** may include a constant diameter **15** along the longitudinal extent of the attachment section **14** from the first terminal end **17** to the guide section **20** in some embodiments such that the attachment section **14** may be selectively inserted a desired distance into a bur mount **82** located in a head **84** of the dental drill **80**. In some embodiments, the diameter **15** of the attachment section **14** is in the range of 2.0 mm to 3.0 mm, although other diameters greater or less than these values may be utilized based on the requirements of the teeth being modified by the bur **10** and the requirements of the drill **80**. In other embodiments, as shown in FIGS. 1 and 4, the end of the attachment section **14** of the shank **12** has a taper **13** as the end that extends into the guide section **20**.

(17) The guide section **20** is located between the attachment section **14** and the second terminal end

18 of the shank **12**, as shown in FIG. **1**. The guide section **20** and the attachment section **14** may be formed of metallic materials. For example, the guide section **20** and attachment **14** may be formed zirconia, zirconia metal-oxide, zirconium, and the like. The guide section **20** includes at least one portion having a first width measured in a transverse direction of the shank and a second portion having a second width measured in the transverse direction, the first width being different than the second width.

(18) For example, in the illustrative embodiment, the guide section **20** includes an inner end **21** located adjacent the attachment section **14** and an outer end **22** located adjacent the head **30**, as shown in FIG. **1**. The at least one portion having the first width **36** is located at the inner end **21** of the guide section **20** and the second portion having the second width **37** is located at the outer end **22** of the guide section **20**, the first width **36** being greater than the second width **37**. As a result, the guide section **20** in some embodiments has a constant first taper from the at least one portion having the first width **36** to the second portion having the second width **37** so as to form a frustum-shape. The width **38** at a midpoint of the guide section **20** is halfway between the first width **36** and the second width **37**. In some embodiments, the diameter **15** of the attachment portion **14** of the shank **12** is equal to the first width **36**. This shape of the guide section **20** allows for precision application of the head to a tooth for minimal and precise tooth material removal without the guide section removing tooth material when coming into contact with adjacent teeth.

(19) The head **30** is arranged on the second terminal end **18** of the shank **12** and includes a tip **32** located opposite the shank **12**, as shown in FIGS. **1-8**. The tip **32** includes at least one rounded portion configured to remove portions of the tooth. For example, as shown in FIG. **1** and as shown in greater detail in FIG. **3**, the at least one rounded portion of the tip **32** is a semi-spherical shape. In the illustrative embodiment, the semi-spherical shape of the tip **32** has a constant radius of curvature around the entire tip **32**. In other embodiments, the semi-spherical shape of the tip **32** may be nonconstant such that the tip includes small grooves and bumps around the tip **32**.

(20) In the illustrative embodiment, the head **30** is entirely coated in an abrasive diamond material **35**, as shown in FIGS. **1** and **3**. The abrasive diamond material **35** may be adhered to the surface of the head **30**. The diamond material **35** is extremely hard and, when the bur **10** is rotated at high speeds via the dental drill **80**, is able to grind down the tooth tissue which typically consists of enamel, dentin, cementum, and restorative materials. The diamonds of the abrasive diamond material **35** may be in the range from ultra-fine to very coarse.

(21) The head **30** also includes an inner terminal end **33** located adjacent the second terminal end **18** of the shank **12** and a main body portion **34** located between the inner terminal end **33** and the tip **32**, as shown in FIG. **3**. In the illustrative embodiment, the inner terminal end **33** has a third width **39** that is equal to the second width **37** of the second portion of the guide section **20** of the shank **12**. An inner end **41** of the tip **32** located adjacent the main body portion **34** of the head **30** has a fourth width **43** that is less than the third width **39**. Specifically, in some embodiments, the main body portion **34** of the head has a constant second taper from the inner terminal end **33** having the third width **39** to the inner end **41** of the tip **32** having the fourth width **43** so as to form a frustum-shape.

(22) A taper angle of the second taper is equal to a taper angle of the first taper of the guide section **20** of the shank **12** such that the shank **12** and the main body portion **34** of the head **30** have a constant taper from the inner end **21** of the guide section **20** to the inner end **41** of the tip **32**, as can be seen in FIGS. **1** and **8**. In some embodiments, the width of the tip **32** at the inner end **41** is about 1.5 mm. The length of the head **30**, from inner terminal end **33** to the outermost end of the tip **32** is about 2.0 mm in some embodiments. In other embodiments, the width of the tip **32** may be in the range of 1.0 mm to 2.0 mm, and the length may be in the range of 1.5 mm to 3.0 mm. These ranges may be adjusted based on the needs of the dental procedure and the teeth being modified.

(23) In the illustrative embodiment, the guide section **20** of the shank **12** and the main body portion **34** of the head **30** are generally cylindrical such that the first width **36**, the second width **37**, the

third width **39**, and fourth width **43** are diameters of the generally cylindrical shank **12** and head **30**. (24) Exemplary embodiments of the dental bur **10** in operation are shown in FIGS. **2** and **5-7**. The dental bur **10** may be inserted into an area **90** of the patient's mouth for grinding down of tooth material minimally and precisely for the purposes of inserting a crown, veneers, or other similar procedure. Typically, before the crown or veneer is inserted and placed over the tooth, a portion of the outer surface of the tooth, usually the enamel, is grinded down to create an even, consistent surface against which to adhere the crown or veneer. As can be seen in FIG. **2**, the dental bur **10** may be inserted into a lower area of a tooth or between adjacent teeth **92**, **94** in order to begin removal of tooth material of one of the teeth **92**, **94** or both teeth **92**, **94**. Because the only portion of the dental bur **10** that includes coarse, abrasive material capable of removing tooth material is the head **30**, the shank **12** can come into contact with upper portions of the teeth **92**, **94** without removing tooth material. This is useful in situations where removal of tooth material is required in difficult to reach areas and accidental removal of tooth material is undesired in areas where the shank **12** may bump or accidentally contact the tooth or adjacent teeth.

(25) Another embodiment of a dental bur **110** in accordance with the present disclosure is shown in FIGS. **9** and **10**. The dental bur **110** is similar to the dental bur **10** shown in FIGS. **1-8** and described herein. Accordingly, similar reference numbers in the **100** series indicate features that are common between the dental bur **110** and the dental bur **10**. The description of the dental bur **10** is incorporated by reference to apply to the dental bur **110**, except in instances when it conflicts with the specific description and the drawings of the dental bur **110**.

(26) Similar to the dental bur **10**, the dental bur **110** of the present embodiment includes a shank **112** and a head **130**, as shown in FIGS. **9** and **10**. The shank **112** includes an attachment section **114** that is configured to be arranged within the bur mount **82** of the dental drill **80**, and a guide section **120** located between the attachment section **114** and a second terminal end **118** of the shank **112**. The guide section **120** includes at least one portion having a first width measured in a transverse direction of the shank and a second portion having a second width measured in the transverse direction, the first width being different than the second width.

(27) For example, in the illustrative embodiment, the guide section **120** includes an inner end **121** located adjacent the attachment section **114** and an outer end **122** located adjacent the head **130**, as shown in FIG. **9**. The at least one portion having the first width **136** is located at the inner end **121** of the guide section **120** and the second portion having the second width **137** is located at the outer end **122** of the guide section **120**. The first width **136** is greater than the second width **137** such that the guide section **120** in some embodiments has a constant first taper from the at least one portion having the first width **136** to the second portion having the second width **137** so as to form a frustum-shape. The width **138** at a midpoint of the guide section **120** is halfway between the first width **136** and the second width **137**. In some embodiments, the first width **136** of the inner end **121** is equal to the diameter of the attachment section **114**. This shape of the guide section **120** allows for precision application of the head to a tooth for minimal and precise tooth material removal without the guide section removing tooth material when coming into contact with adjacent teeth.

(28) The head **130** is arranged on the second terminal end **118** of the shank **112** and includes a tip **132** located opposite the shank **112**, as shown in FIGS. **9** and **10**. The tip **132** includes at least one rounded portion configured to remove portions of the tooth. For example, as shown in FIG. **9** and as shown in greater detail in FIG. **10**, the at least one rounded portion of the tip **132** is an annular rounded ridge **131** that extends around the tip **132** at the inner end **141** of the tip **132**. A distal end **145** of the tip **132** located opposite the inner end **141** of the tip **132** is an annular flat surface **147** connected to the annular rounded ridge **131** so as to form a rounded chamfered shape. This chamfered shape, having the flat top surface and rounded edges may also be configured as a modified shoulder. In the illustrative embodiment, the head **130** is entirely coated in an abrasive diamond material **135**, as shown in FIGS. **1** and **3**. The abrasive diamond material **135** may be

adhered to the surface of the head **130**. The diamonds of the abrasive diamond material **135** may be in the range from ultra-fine to very coarse.

(29) The head **130** also includes an inner terminal end **133** located adjacent the second terminal end **118** of the shank **112** and a main body portion **134** located between the inner terminal end **133** and the tip **132**, as shown in FIG. **10**. In the illustrative embodiment, the inner terminal end **133** has a third width **139** that is equal to the second width **137** of the second portion of the guide section **120** of the shank **112**. The inner end **141** of the tip **132** is located adjacent the main body portion **134** of the head **130** and has a fourth width **143** that is less than the third width **139**. Specifically, in some embodiments, the main body portion **134** of the head has a constant second taper from the inner terminal end **133** having the third width **139** to the inner end **141** of the tip **132** having the fourth width **143** so as to form a frustum-shape.

(30) A taper angle of the second taper is equal to a taper angle of the first taper of the guide section **120** of the shank **112** such that the shank **112** and the main body portion **134** of the head **130** have a constant taper from the inner end **121** of the guide section **120** to the inner end **141** of the tip **132**, as can be seen in FIGS. **9** and **10**. In some embodiments, the width of the tip **132** at the inner end **141** is about 1.5 mm. The length of the head **130**, from inner terminal end **133** to the distal end **145** of the tip **132** is about 2.0 mm in some embodiments. In other embodiments, the width of the tip **132** may be in the range of 1.0 mm to 2.0 mm, and the length may be in the range of 1.5 mm to 3.0 mm. These ranges may be adjusted based on the needs of the dental procedure and the teeth being modified.

(31) Another embodiment of a dental bur **210** in accordance with the present disclosure is shown in FIG. **11**. The dental bur **210** is similar to the dental burs **10**, **110** shown in FIGS. **1-10** and described herein. Accordingly, similar reference numbers in the **200** series indicate features that are common between the dental bur **210** and the dental burs **10**, **110**. The descriptions of the dental burs **10**, **110** are incorporated by reference to apply to the dental bur **210**, except in instances when they conflict with the specific description and the drawings of the dental bur **210**.

(32) Similar to the dental burs **10**, **110**, the dental bur **210** of the present embodiment includes a shank **212** and a head **230**, as shown in FIG. **11**. The shank **212** includes an attachment section **214** that is configured to be arranged within the bur mount **82** of the dental drill **80**, and a guide section **220** located between the attachment section **214** and a second terminal end **218** of the shank **212**.

(33) In the illustrative embodiment, the guide section **220** includes an inner end **221** located adjacent the attachment section **214** and an outer end **222** located adjacent the head **230**, and an intermediate section **224** located longitudinally between the inner end **221** and the outer end **222**, as shown in FIG. **11**. The at least one portion having the first width **236** is located at the inner end **221** of the guide section **220** and the second portion having the second width **237** is located at the outer end **222** of the guide section **220**. A third portion of the guide section is located between the at least one portion and the second portion within the intermediate section **224** and has a third width **238** measured in the transverse direction. The first width **236** and the second width **237** are greater than the third width **238** such that the guide section **220** has an hourglass shape. In the illustrative embodiment, the second width **237** is smaller than the first width **236**, as shown in FIG. **11**. This shape of the guide section **220** allows for precision application of the head to a tooth for minimal and precise tooth material removal without the guide section removing tooth material when coming into contact with adjacent teeth. In some embodiments, the first width **236** of the inner end **221** is equal to the diameter of the attachment section **214**.

(34) The head **230** is arranged on the second terminal end **218** of the shank **212** and includes a tip **232** located opposite the shank **212**, as shown in FIG. **11**. The tip **232** includes at least one rounded portion configured to remove portions of the tooth. For example, as shown in FIG. **11**, the at least one rounded portion of the tip **232** is an annular rounded ridge **231** that extends around the tip **232** at the inner end **241** of the tip **232**. A distal end **245** of the tip **232** located opposite the inner end **241** of the tip **232** is an annular flat surface **247** connected to the annular rounded ridge **231** so as to

form a rounded chamfered shape. This chamfered shape, having the flat top surface and rounded edges may also be configured as a modified shoulder. In other embodiments, the at least one rounded portion of the tip **232** may be formed as a semi-spherical shape having a constant or non-constant radius of curvature, as shown in FIGS. **1-8**. In the illustrative embodiment, the head **230** is entirely coated in an abrasive diamond material **235**. The abrasive diamond material **235** may be adhered to the surface of the head **230**. The diamonds of the abrasive diamond material **235** may be in the range from ultra-fine to very coarse.

(35) The head **230** also includes an inner terminal end **233** located adjacent the second terminal end **218** of the shank **212** and a main body portion **234** located between the inner terminal end **233** and the tip **232**, as shown in FIG. **11**. In the illustrative embodiment, the inner terminal end **233** has a third width **239** that is equal to the second width **237** of the second portion of the guide section **220** of the shank **212**. The inner end **241** of the tip **232** is located adjacent the main body portion **234** of the head **230** and has a fourth width **243** that is less than the third width **239**. In some embodiments, the width of the tip **232** at the inner end **241** is about 1.5 mm. The length of the head **230**, from inner terminal end **233** to the distal end **245** of the tip **232** is about 2.0 mm in some embodiments. In other embodiments, the width of the tip **232** may be in the range of 1.0 mm to 2.0 mm, and the length may be in the range of 1.5 mm to 3.0 mm. These ranges may be adjusted based on the needs of the dental procedure and the teeth being modified.

(36) Another embodiment of a dental bur **310** in accordance with the present disclosure is shown in FIG. **12**. The dental bur **310** is similar to the dental burs **10**, **110**, **210** shown in FIGS. **1-11** and described herein. Accordingly, similar reference numbers in the **300** series indicate features that are common between the dental bur **310** and the dental burs **10**, **110**, **210**. The descriptions of the dental burs **10**, **110**, **210** are incorporated by reference to apply to the dental bur **310**, except in instances when they conflict with the specific description and the drawings of the dental bur **310**.

(37) Similar to the dental burs **10**, **110**, **210**, the dental bur **310** of the present embodiment includes a shank **312** and a head **330**, as shown in FIG. **12**. The shank **312** includes an attachment section **314** that is configured to be arranged within the bur mount **82** of the dental drill **80**, and a guide section **320** located between the attachment section **314** and a second terminal end **318** of the shank **312**.

(38) In the illustrative embodiment, the guide section **320** includes an inner end **321** located adjacent the attachment section **314** and an outer end **322** located adjacent the head **330**, and an intermediate section **324** located longitudinally between the inner end **321** and the outer end **322**, as shown in FIG. **12**. The at least one portion having the first width **336** is located at the inner end **321** of the guide section **320** and the second portion having the second width **337** is located at the outer end **322** of the guide section **320**. A third portion of the guide section is located between the at least one portion and the second portion within the intermediate section **324** and has a third width **338** measured in the transverse direction. The first width **336**, the second width **337**, and the third width **338** have an equal diameter such that the guide section **320** has a straight, cylindrical shape. This shape of the guide section **320** allows for precision application of the head to a tooth for minimal and precise tooth material removal without the guide section removing tooth material when coming into contact with adjacent teeth. In some embodiments, the first width **336** of the inner end **321** is equal to the diameter of the attachment section **314**.

(39) The head **330** is arranged on the second terminal end **318** of the shank **312** and includes a tip **332** located opposite the shank **312**, as shown in FIG. **12**. The tip **332** includes at least one rounded portion configured to remove portions of the tooth. For example, as shown in FIG. **12**, the at least one rounded portion of the tip **332** is an annular rounded ridge **331** that extends around the tip **332** at the inner end **341** of the tip **332**. A distal end **345** of the tip **332** located opposite the inner end **341** of the tip **332** is an annular flat surface **347** connected to the annular rounded ridge **231** so as to form a rounded chamfered shape. This chamfered shape, having the flat top surface and rounded edges may also be configured as a modified shoulder. In other embodiments, the at least one

rounded portion of the tip **332** may be formed as a semi-spherical shape having a constant or non-constant radius of curvature, as shown in FIGS. **1-8**. In the illustrative embodiment, the head **330** is entirely coated in an abrasive diamond material **335**. The abrasive diamond material **335** may be adhered to the surface of the head **330**. The diamonds of the abrasive diamond material **335** may be in the range from ultra-fine to very coarse.

(40) The head **330** also includes an inner terminal end **333** located adjacent the second terminal end **318** of the shank **312** and a main body portion **334** located between the inner terminal end **333** and the tip **332**, as shown in FIG. **12**. In the illustrative embodiment, the inner terminal end **333** has a third width **339** that is equal to the second width **337** of the second portion of the guide section **320** of the shank **312**. The inner end **341** of the tip **332** is located adjacent the main body portion **334** of the head **330** and has a fourth width **343** that is less than the third width **339**. In some embodiments, the width of the tip **332** at the inner end **341** is about 1.5 mm. The length of the head **330**, from inner terminal end **333** to the distal end **345** of the tip **332** is about 2.0 mm in some embodiments. In other embodiments, the width of the tip **332** may be in the range of 1.0 mm to 2.0 mm, and the length may be in the range of 1.5 mm to 3.0 mm. These ranges may be adjusted based on the needs of the dental procedure and the teeth being modified.

(41) While the disclosure has been illustrated and described in detail in the foregoing drawings and description, the same is to be considered as exemplary and not restrictive in character, it being understood that only illustrative embodiments thereof have been shown and described and that all changes and modifications that come within the spirit of the disclosure are desired to be protected.

Claims

1. A dental bur for removing precise portions of a tooth, comprising: a shank having a first terminal end, a second terminal end located opposite the first terminal end in a longitudinal direction of the shank, an attachment section located adjacent the first terminal end, and a guide section located between the attachment section and the second terminal end, the attachment section being configured to be inserted into and rotationally driven by a high-speed rotary device, the guide section including at least one portion having a first portion having a first width measured in a transverse direction of the shank and a second portion having a second width measured in the transverse direction, and the first width being different than the second width; and a head arranged on the second terminal end of the shank and including a main body portion arranged on the second terminal end of the shank and a tip located on an outer terminal end of the main body portion and opposite the shank, the tip having at least one rounded portion configured to remove portions of the tooth, and the head being entirely coated in an abrasive diamond material, wherein the main body portion is cylindrical and continuously tapered from an inner terminal end of the main body portion to the tip, the main body portion tapering from a larger width to a smaller width in a direction from the shank to the tip, wherein the at least one rounded portion of the tip has a semi-spherical shape, wherein the guide section is devoid of abrasive diamond material and devoid of an abrasive coating, wherein the guide section includes an inner end located adjacent the attachment section, an outer end located adjacent the head, and an intermediate section located between the inner end and the outer end, the first portion having the first width is located at the inner end of the guide section, the second portion having the second width is located at the outer end of the guide section, a third portion of the guide section is located between the first portion and the second portion within the intermediate section and has a third width measured in the transverse direction, and the first width and the second width are greater than the third width such that the guide section has an hourglass shape, wherein the third width is the smallest width along a longitudinal length of the guide section, and wherein a first lengthwise location that is located generally centrally along the longitudinal length of the guide section has the third width.

2. The dental bur of claim 1, wherein the first width is greater than the second width.

3. The dental bur of claim 2, wherein the inner terminal end of the main body portion has a fourth width that is equal to the second width of the shank at the second terminal end of the shank.
4. The dental bur of claim 3, wherein an inner end of the tip located adjacent the main body portion of the head has a fifth width that is less than the fourth width.
5. The dental bur of claim 4, wherein the guide section of the shank and the main body portion of the head define cross sections at each of the first, second, third, fourth, and fifth widths that are generally circular such that the first, second, third, fourth, and fifth widths are diameters of the shank and head.
6. A high-speed dental drill, comprising: a drill head; a bur mount arranged within the drill head and configured to rotate at high speeds; a motor for rotating the bur mount; a dental bur arranged within the bur mount, the dental bur including: a shank having a first terminal end, a second terminal end located opposite the first terminal end in a longitudinal direction of the shank, an attachment section located adjacent the first terminal end, and a guide section located between the attachment section and the second terminal end, the attachment section being configured to be inserted into and rotationally driven by a high-speed rotary device, the guide section including at least one portion having a first portion having a first width measured in a transverse direction of the shank and a second portion having a second width measured in the transverse direction, and the first width being different than the second width; and a head arranged on the second terminal end of the shank and including a tip located opposite the shank, the tip having at least one rounded portion configured to remove portions of the tooth, and the head being entirely coated in an abrasive diamond material, wherein the head includes a main body portion extending between an inner terminal end located adjacent the second terminal end of the shank and an outer terminal end located adjacent the tip, wherein the inner terminal end has a third width measured in a transverse direction of the head and the second terminal end of the shank has a fourth width, wherein the third width is equal to the fourth width of the second terminal end of the shank, wherein a width of the main body portion of the head at every position between the inner terminal end and the outer terminal end is less than or equal to the third and fourth widths and is greater than or equal to a width of the tip at every position between an inner end of the tip adjacent the main body portion and a distal end of the tip, and wherein the guide section is devoid of abrasive diamond material and devoid of an abrasive coating, wherein the guide section includes an inner end located adjacent the attachment section and an outer end located adjacent the head, the first portion having the first width is located at the inner end of the guide section, the second portion having the second width is located at the outer end of the guide section, and the first width is greater than the second width, wherein the guide section has a constant first taper from the first portion having the first width to the second portion having the second width so as to form a frustum-shape, wherein the at least one rounded portion of the tip has a semi-spherical shape, wherein the inner end of the tip located adjacent the main body portion of the head has a fifth width that is less than the third and fourth widths, and wherein the main body portion of the head has a constant second taper from the inner terminal end having the third width to the inner end of the tip having the fifth width so as to form a frustum-shape, and a taper angle of the second taper is equal to a taper angle of the constant first taper of the guide section of the shank.
7. A high-speed dental drill, comprising: a drill head; a bur mount arranged within the drill head and configured to rotate at high speeds; a motor for rotating the bur mount; a dental bur arranged within the bur mount, the dental bur including: a shank having a first terminal end, a second terminal end located opposite the first terminal end in a longitudinal direction of the shank, an attachment section located adjacent the first terminal end, and a guide section located between the attachment section and the second terminal end, the attachment section being configured to be inserted into and rotationally driven by a high-speed rotary device, the guide section including at least one portion having a first portion having a first width measured in a transverse direction of the shank and a second portion having a second width measured in the transverse direction, and the

first width being different than the second width; and a head arranged on the second terminal end of the shank and including a tip located opposite the shank, the tip having at least one rounded portion configured to remove portions of the tooth, and the head being entirely coated in an abrasive diamond material, wherein the head includes a main body portion extending between an inner terminal end located adjacent the second terminal end of the shank and an outer terminal end located adjacent the tip, wherein the inner terminal end has a third width measured in a transverse direction of the head and the second terminal end of the shank has a fourth width, wherein the third width is equal to the fourth width of the second terminal end of the shank, wherein a width of the main body portion of the head at every position between the inner terminal end and the outer terminal end is less than or equal to the third and fourth widths and is greater than or equal to a width of the tip at every position between an inner end of the tip adjacent the main body portion and a distal end of the tip, wherein the guide section includes a constant first taper between the attachment section and the second terminal end, the guide section being devoid of abrasive diamond material and devoid of an abrasive coating, and wherein the main body portion of the head has a constant second taper from the inner terminal end having the third width to the inner end of the tip so as to form a frustum-shape, and a taper angle of the second taper is equal to a taper angle of the first taper of the guide section of the shank.

8. The high-speed dental drill of claim 7, wherein the guide section includes an inner end located adjacent the attachment section and an outer end located adjacent the head, the first portion having the first width is located at the inner end of the guide section, the second portion having the second width is located at the outer end of the guide section, and the first width is greater than the second width.

9. The high-speed dental drill of claim 8, wherein the constant first taper of the guide section extends from the first portion having the first width to the second portion having the second width so as to form a frustum-shape.

10. The high-speed dental drill of claim 9, wherein the at least one rounded portion of the tip has a semi-spherical shape.

11. The high-speed dental drill of claim 9, wherein the at least one rounded portion of the tip is an annular rounded ridge that extends around the tip at the inner end of the tip, and wherein the distal end of the tip located opposite the inner end of the tip is an annular flat surface connected to the annular rounded ridge.
